أ. سعاد حامد عز الحاوي

التكامل Integration

* التكامل غير المحدود Indefinite Integral

تعريف :- إذا كانت f معرفة على الفترة $[a,b]$ ، وإذا وجدت دالة F ، حيث F متصلة على $[a,b]$ ، وقابلة للاشتقاق على (a,b) و $F'(x)=f(x)$ ، فإن F تسمى العلاقة العكسية لتفاضل الدالة f أو تكامل f غير المحدود ، ويرمز لذلك بالرمز

$$F(x)=\int f(x)\,dx$$

* قواعد لإيجاد بعض التكاملات :-

① إذا كانت $f(x)=x^n$ حيث $n\in\mathbb{R}$ ، $n\neq -1$ فإن

(i) $$\int x^n\,dx=\frac{x^{n+1}}{n+1}+c$$

(ii) $$\int f'(x)\left[f(x)\right]^n dx=\frac{\left[f(x)\right]^{n+1}}{n+1}+C$$

② إذا كانت f ، g دالتين قابلتين للتكامل ، وكان k_1 ، k_2 مقدارين ثابتين فإن

$$\int [k_1 f(x) + k_2 g(x)]\,dx = k_1 \int f(x)\,dx + k_2 \int g(x)\,dx$$

3) إذا كانت $n=-1$

$$\int f'(x)\,[\overset{u}{f(x)}]\,dx = \int \frac{f'(x)}{[f(x)]}\,dx = \ln|f(x)| + C$$

أمثلة

Ex ① $\int x^{10}\,dx = \dfrac{x^{10+1}}{10+1} + C$

$$= \frac{1}{11} x^{11} + C$$

② $\int (3x^3 + 2x^2)\,dx = 3\int x^3\,dx + 2\int x^2\,dx$

$$= \frac{3}{4} x^4 + \frac{2}{3} x^3 + C$$

③ $\int x\,(x^2+4)^2\,dx = \frac{1}{2}\int 2x\,(x^2+4)^2\,dx$

$$= \frac{1}{2} * \frac{(x^2+4)^3}{3} + C$$

$$= \frac{1}{6}(x^2+4)^3 + C$$

④ $\int \dfrac{x}{(x^2+4)^3}\,dx = \int x\,(x^2+4)^{-3}\,dx$

$$= \frac{1}{2}\int 2x\,(x^2+4)^{-3}\,dx$$

$$\therefore \int \frac{x}{(x^2+4)^3}\,dx = \frac{1}{2} * \frac{(x^2+4)^{-2}}{-2} + C$$

$$= -\frac{1}{4} * \frac{1}{(x^2+4)^2} + C$$

$$= -\frac{1}{4(x^2+4)^2} + C$$

(5) $$\int (x^2+1)\sqrt{x^3+3x}\,dx = \frac{1}{3}\int 3(x^2+1)\,(x^3+3x)^{\frac{1}{2}}\,dx$$

$$= \frac{1}{3} * \frac{2}{3}(x^3+3x)^{\frac{3}{2}} + C$$

$$= \frac{2}{9}(x^3+3x)^{3/2} + C$$

(6) $$\int \cos^3 x \sin x\,dx \quad \Big\} \quad \because \underset{\text{[illegible]}}{(\cos x)'} = -\sin x$$

$$= -\int -\sin x \cos^3 x\,dx$$

$$= -\frac{\cos^4 x}{4} + C$$

(7) $$\int (1+e^{2x})^{-5} e^{2x}\,dx \quad \Big\} \quad \because (1+e^{2x})' = 2e^{2x}$$

$$= \frac{1}{2} * \frac{(1+e^{2x})^{-4}}{-4} + C$$

$$= -\frac{1}{8}(1+e^{2x})^{-4} + C$$

(8) $\displaystyle\int \frac{\sec^2 x}{\tan x}\,dx$ $\quad\Big\}\quad (\tan x)' = \sec^2 x$

$= \ln|\tan x| + C$

(9) $\displaystyle\int \frac{e^{3x}}{e^{3x}+10}\,dx$ $\quad\Big\}\quad (e^{3x})' = 3e^{3x}$

$= \displaystyle\frac{1}{3}\int \frac{3e^{3x}}{e^{3x}+10}\,dx = \frac{1}{3}\ln|e^{3x}+10| + C$

(10) $\displaystyle\int \left(x^4 + \frac{5}{x}\right)dx = \frac{x^5}{5} + 5\ln|x| + C$

(11) $\displaystyle\int \frac{dx}{\sqrt[3]{2x+1}} = \int (2x+1)^{-\frac{1}{3}}\,dx = \frac{1}{2}\int 2\,(2x+1)^{-\frac{1}{3}}\,dx$

$= \frac{1}{2}\cdot\frac{3}{2}(2x+1)^{\frac{2}{3}} + C$

$= \frac{3}{4}(2x+1)^{2/3} + C$

(12) $\int \left[\frac{4}{x^3} - x(x^2-1)^4\right] dx = \int 4x^{-3} dx - \int x(x^2-1)^4 dx$

$= \cancel{4}^{2} * \frac{x^{-2}}{\cancel{-2}_{1}} - \frac{1}{2}\int 2x(x^2-1)^4 dx$

$= -\frac{2}{x^2} - \frac{1}{2} * \frac{(x^2-1)^5}{5} + C$

$= \frac{-2}{x^2} - \frac{1}{10}(x^2-1)^5 + C$

(13) $\int \sin(3x-1)\cos^2(3x-1)\,dx$

$\left[\cos(3x-1)\right]' = -3\sin(3x-1)$

$= -\frac{1}{3}\int -3\sin(3x-1)\cos^2(3x-1)\,dx$

$= -\frac{1}{3} * \frac{\cos^3(3x-1)}{3} + C$

$= -\frac{1}{9}\cos^3(3x-1) + C$

(14) $\int \frac{1}{\sqrt[5]{x-1}}\,dx = \int (x-1)^{-\frac{1}{5}} dx$

$= \frac{5}{4}(x-1)^{4/5} + C$

* تكامل الدوال المثلثية

مراجعة / إشتقاق الدوال المثلثية

$f(x)$	$f'(x)$
$\sin x$	$\cos x$
$\cos x$	$-\sin x$
$\tan x$	$\sec^2 x$
$\sec x$	$\sec x \tan x$
$\csc x$	$-\csc x \cot x$
$\cot x$	$-\csc^2 x$

تكامل الدوال المثلثية :-

$f(x)$	$\int f(x)\,dx$
$\sin x$	$-\cos x + C$
$\cos x$	$\sin x + C$
$\tan x$	$\ln\lvert\sec x\rvert + C$ $\quad -\ln\lvert\cos x\rvert + C$
$\sec x$	$\ln\lvert\sec x + \tan x\rvert + C$
$\csc x$	$\ln\lvert\csc x - \cot x\rvert + C$
$\cot x$	$\ln\lvert\sin x\rvert + C$ $\quad -\ln\lvert\csc x\rvert + C$
$\sec^2 x$	$\tan x + C$
$\csc^2 x$	$-\cot x + C$
$\sec x \tan x$	$\sec x + C$
$\csc x \cot x$	$-\csc x + C$

أمثلة :-

① $\int (\sin x + x^2)\,dx = \int \sin x\,dx + \int x^2\,dx$

$= -\cos x + \frac{x^3}{3} + C$

② $\int 5 \tan x\,dx = 5\int \tan x\,dx = 5 \ln|\sec x| + C$

③ $\int \cos(4x)\,dx = \frac{1}{4}\int 4\cos(4x)\,dx$

$= \frac{1}{4} \sin(4x) + C$

④ $\int (2x+7)\tan(x^2+7x)\,dx = \ln|\sec(x^2+7x)| + C$

⑤ $\int \sin(3x-1)\,dx = \frac{1}{3}\int 3\sin(3x-1)\,dx$

$= -\frac{1}{3}\cos(3x-1) + C$

⑥ $\int \sec^2\left(\frac{x}{2}\right)dx = 2\int \frac{1}{2}\sec^2\left(\frac{x}{2}\right)dx$

$= 2\tan\left(\frac{x}{2}\right) + C$

⑦ $\int \frac{\cos\sqrt{x}}{\sqrt{x}}\,dx = 2\int \frac{\cos\sqrt{x}}{2\sqrt{x}}\,dx$

$= 2\sin\sqrt{x} + C$

(8) $\int t\,(4-2t^2)^7\,dt = -\frac{1}{4}\int 4t\,(4-2t^2)^7\,dt$

$= -\frac{1}{4} * \frac{(4-2t^2)^8}{8} + C$

$= -\frac{1}{32}\,(4-2t^2)^8 + C$

(9) $\int \frac{\cos 3x}{\sin^2 3x}\,dx = \int \cos 3x\,\sin^{-2} 3x\,dx$

$= \frac{1}{3}\int 3\cos 3x\,\sin^{-2} 3x\,dx$

$= \frac{1}{3} * \frac{\sin^{-1} 3x}{-1} + C$

$= -\frac{1}{3\sin 3x} + C$

$= -\frac{1}{3}\csc 3x + C$

(10) $\int \left(\frac{\sec x}{1+\tan x}\right)^2 dx = \int \frac{\sec^2 x}{(1+\tan x)^2}\,dx$

$= \int \sec^2 x\,(1+\tan x)^{-2}\,dx$

$= \frac{(1+\tan x)^{-1}}{-1} + C$

$= -\frac{1}{1+\tan x} + C$

(11) $\int x^2 \csc^2 x^3\,dx$

$$\therefore \int x^2 \csc^2 x^3 \, dx = \frac{1}{3}\int 3x^2 \csc^2 x^3 \, dx$$

$$= -\frac{1}{3}\cot x^3 + C$$

H.W

① $\int \frac{\cos x}{\sin^2 x} \, dx$ ② $\int [\csc 7x + \cot 10x] \, d$

③ $\int x \sec(x^2+4) \tan(x^2+4) \, dx$

④ $\int x \frac{\sec^2 4x^2}{\tan 4x^2} \, dx$

* تكامل الدوال الأسية

① $\int f'(x) e^{f(x)} \, dx = \frac{e^{f(x)}}{\ln e} + C$; $\ln e = 1$

$$= e^{f(x)} + C$$

② $\int f'(x) \, a^{f(x)} \, dx = \frac{a^{f(x)}}{\ln a} + C$

أمثلة

① $\int 3^{x}\,dx = \dfrac{3^{x}}{\ln 3} + C$

② $\int e^{x}\,dx = e^{x} + C$

③ $\int (e^{x} + 5^{x} + 3)\,dx = \int e^{x}dx + \int 5^{x}dx + 3x +$

$= e^{x} + \dfrac{5^{x}}{\ln 5} + 3x + C$

④ $\int (3^{x} - 4e^{2x})\,dx = \dfrac{3^{x}}{\ln 3} - 2e^{2x} + C$

⑤ $\int \cos x\, e^{\sin x}\,dx = e^{\sin x} + C$

⑥ $\int \tan x \cos x\, e^{\cos x}\,dx = \int \dfrac{\sin x}{\cancel{\cos x}} \cancel{\cos x}\, e^{\cos x}\,dx$

$= \int \sin x\, e^{\cos x}\,dx$

$= -\int -\sin x\, e^{\cos x}\,dx$

$= -e^{\cos x} + C$

⑦ $\int x(e^{x^2} + 2x)\,dx$

$$= \int x\left(e^{x^2}+2x\right)dx = \int x e^{x^2} dx + 2\int x^2 dx$$

$$= \frac{1}{2}\int 2x e^{x^2} dx + \frac{2}{3}x^3 + C$$

$$= \frac{1}{2} e^{x^2} + \frac{2}{3}x^3 + C$$

H. W

① $\int \sec^2 x \, e^{\tan x} dx$

② $\int (2x-1) \, 2^{(x^2-x)} dx$

③ $\int x \, g^{1-x^2} dx$

④ $\int \left[e^{2+3x} + x \tan x^2 \right] dx$

⑤ $\int \frac{e^{\sqrt{x}}}{\sqrt{x}} dx$